Description

Illustration





Figure caption

Reactive powder concrete is often used in structural applications where achieving a minimum strength within a defined time period is critical. 1. The Delaware Memorial Bridge, USA, where reactive powder concrete was used in renovation of the deck. 2. Nipigon Bridge, Canada where reactive powder concrete was used in specific joint connections. Images used under license from Shutterstock.com

The material

Reactive powder concrete (RPC) is an extremely high strength concrete material. Boasting high compressive strengths, RPCs are also extremely ductile, hundreds of times more than normal concrete, plus possessing extremely good flexural strength. These concretes reach performance levels far beyond conventional and even most high performance concretes, hence they are also known as ultra-high performance concretes (UHPC). The careful grading of particulates, such that maximum compaction is achieved along with the addition of very fine whiskers of steel wire, result in a material that defies the normal behavior of concrete. Large aggregate are not used and the fine sand, Portland cement and the addition of silica fume are carefully proportioned such that their interaction maximizes bonding.
Types of reactive powder concrete are premixed and self-leveling, allowing for placement in formwork without the need for vibration. Also, the very fine particle makeup of the mixture allows for use in complex precast forms. Both dry and wet mixes are sold. The cost of this kind of concrete is currently several times that of normal concrete.
Note that the properties here regard unreinforced concrete, rather than the significantly improved properties that can be achieved by reinforcement with concrete.

Compositional summary

3.5-4.0:1.5:1:6:1:0.1:1:cement (ASTM Type 2):silica fume (microsilica):ground quartz:silica sand:steel fiber:superplasticizer:water (Note: no course aggregate)
Mix constituents and proportions are varied to produce widely ranging properties.

General properties

Density	2.5e3	-	2.75e3	kg/m^3
Price	* 0.128	-	0.498	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	✗

Mechanical properties

Young's modulus	*	28	-	32	GPa
Shear modulus	*	12	-	14	GPa
Bulk modulus	*	25	-	30	GPa
Bending modulus	*	28	-	32	GPa
Poisson's ratio	*	0.22	-	0.25	
Yield strength (elastic limit)	*	5	-	8	MPa
Tensile strength		5	-	8	MPa
Compressive strength		200	-	600	MPa
Bending strength		48	-	52	MPa
Elongation	*	0.003	-	0.004	% strain
Hardness - Vickers		66.7	-	200	HV
Fatigue strength at 10^7 cycles	*	2	-	3.2	MPa
Fracture toughness		2	-	3	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity	*	0.7	-	0.83	m.°C/W
Thermal expansion coefficient	*	7	-	11	µstrain/°C
Specific heat capacity		900	-	1e3	J/kg.°C
Maximum service temperature		573	-	673	°C
Flammability		Non-flammable			
Emissivity		0.63	-	0.97	

Hygro-thermal properties

Water absorption	1.5	-	1.6	%
Water vapor permeability	2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance	Good			

Electrical properties

Electrical conductor or insulator?	Poor insulator

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Castability	3	-	4
Moldability	4	-	5
Formability	1		
Machinability	1	-	2
Weldability	1		
Solder/brazability	1		

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Excellent
Weak alkalis	Acceptable
Strong alkalis	Limited use
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Excellent
Industrial atmosphere	Acceptable
Rural atmosphere	Acceptable
Marine atmosphere	Acceptable

Geo-economic data for principal component

Annual world production, principal component	1.43e10	-	1.58e10	tonne/yr

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.212	-	0.234	kg/kg
Embodied energy (virgin grade production)	* 1.34	-	1.48	MJ/kg
Water Usage	* 3.23	-	3.57	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 38	-	86.4	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 2.48	-	5.66	kg/kg

Recycling and end of life

Recycle	✗
Functional recycle	✗
Recycle fraction in current supply	13 - 14.4 %
Combust for energy recovery	✗
Landfill	✓
Biodegrade	✗
A renewable resource?	✗

Supporting information

Design guidelines

Reactive powder concrete (RPC), also known as "ultra-high strength" concrete, is prepared by replacing the dense aggregate used in normal concrete with quartz powder, silica fume, steel fibres and superplasticising admixtures, producing a low water/powder ratio, highly fluid, mortar-type mix, without sizeable coarse aggregate pieces.

RPC not only has high strength but also has high ductility.

As its compressive strength ranges from 200 MPa upwards, it is outside the design methods of standard codes and so is commonly supplied as a bespoke manufacturer-designed product.

Technical notes

Note that the properties here regard unreinforced RPC, rather than the significantly improved properties that can be achieved by adding bar or fibre reinforcement.

Typical uses

Currently high performance and highly visible applications in which reduction of the overall bulk of the structural concrete is a priority. Pedestrian bridges, urban bus shelters, thin shells and vaults are ideal applications for this highly engineered material.

Tradenames

RPC (HDR Engineering, North America), Ductal (LaFarge, Bouyges,Rhodia)

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
ASTM C1856: Standard Practice for Fabricating and Testing Specimens of Ultra-High Performance Concrete
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206

Further reading

Fernandez, J. (2005) Material Architecture: emergent materials for innovative buildings and ecological construction. Architectural Press, Oxford, UK: pp: 218-220.
Also see: www.pci.com

Links

Producers

Reference